Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Mock Paper Set 3

Paper reference **8MA0/21**

Mathematics

Advanced Subsidiary
PAPER 21: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 5 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

Pearson

1 A boat company organises a trip to see whales each day.
The company advertises that customers see whales on 80% of its trips.
Throughout the year, Ali goes on 10 of these trips on randomly chosen days.

(a) Suggest a suitable distribution to model the number of trips where Ali sees whales.

(1)

(b) Using this model, find the probability that Ali sees whales on

(i) exactly 8 of the trips,
(ii) at least 7 of the trips.

(3)

Sam really wants to see whales, so goes on a trip each day for 10 consecutive days.

(c) State a limitation of using the model in part (a) to represent the number of trips where Sam sees whales.

(1)

Question 1 continued

(Total for Question 1 is 5 marks)

2 Amelie is a basketball player.
In one season, she scores from 40% of her free throws.

At the end of the season her sports psychologist retires.

Before the start of the next season, Amelie trains with a new sports psychologist.
After training with the new sports psychologist, Amelie takes 60 free throws and scores from 31 of them.

Amelie believes that training with the new sports psychologist has changed the proportion of her free throws from which she scores.

(a) Using a suitable test, assess Amelie's belief.

You should

- state your hypotheses and conclusion clearly
- use a 5% significance level

(4)

(b) Find the *p*-value for the test in part (a).

(1)

Question 2 continued

(Total for Question 2 is 5 marks)

3 A machine produces bolts for ceiling fans.
The machine is set to produce bolts of length 26 mm.
For a bolt to be usable, its length must be (26 ± 2) mm.

A sample of 100 bolts was taken.
The bolts were measured and the results summarised in the table below

Length of bolt (b mm)	Midpoint (x mm)	Frequency
$21 < b \leqslant 22$	21.5	1
$22 < b \leqslant 23$	22.5	5
$23 < b \leqslant 24$	23.5	8
$24 < b \leqslant 25$	24.5	15
$25 < b \leqslant 26$	25.5	31
$26 < b \leqslant 27$	26.5	24
$27 < b \leqslant 28$	27.5	12
$28 < b \leqslant 29$	28.5	4

(a) Estimate the proportion of bolts in this sample that are usable.

(1)

(b) Use linear interpolation to find an estimate for the median bolt length.

(2)

Given that $\sum x = 2560 \quad \sum x^2 = 65\ 751$

(c) find an estimate for

(i) the mean bolt length,
(ii) the standard deviation of the bolt lengths.

(3)

Question 3 continued

(Total for Question 3 is 6 marks)

4 Huan plays a game using a fair 8-sided spinner, numbered 1 to 8
Huan spins the spinner until the game stops.

The game stops according to the following rules

- on the first spin, if the spinner lands on a **square** number, the game stops
- on the second spin, if the spinner lands on an **even** number, the game stops
- on the third spin, if the spinner lands on a **prime** number, the game stops
- if the game is still going after 3 spins, one final spin is taken and the game stops

The random variable S represents the number of times the spinner is spun until the game stops.

(a) Show that $P(S = 2) = \frac{3}{8}$

(2)

(b) Complete the probability distribution table below for the random variable S

(2)

s	1	2	3	4
$P(S = s)$		$\frac{3}{8}$		

(c) Find $P(1.6 < S < 3.2)$

(1)

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Question 4 continued

(Total for Question 4 is 5 marks)

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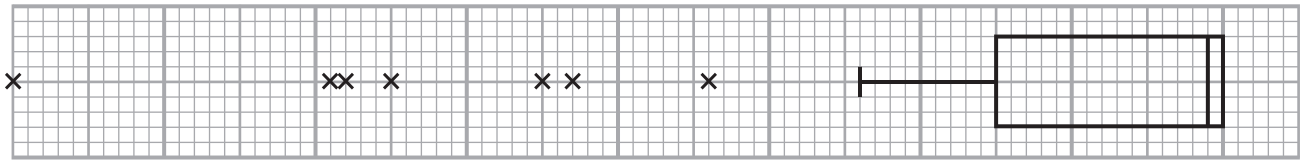
5



Luke finds the diagram above, showing a box plot with no axis.
The diagram lies next to a set of data labelled 'Large data set: Daily Total Rainfall – Hurn 1987'.

Luke thinks that the diagram should be rotated through 180° for it to represent these data.

(a) Using your knowledge of the large data set, explain why Luke is correct.

(1)

A sample of data for the Daily Mean Visibility in Leeming in 1987 was taken and is summarised in the table below

Lower Quartile (dam)	Upper Quartile (dam)
1000	x

An outlier is defined as any value
more than 1.5 × the interquartile range **above** the upper quartile
or more than 1.5 × the interquartile range **below** the lower quartile

The last 6 values of the ordered data are 3300, 3400, 3400, 3600, 3600, 3700

Luke believes that there are 5 outliers at the upper end of the data.

(b) Find the range of possible values for x if Luke is correct.

(4)

The first 6 values of the ordered data are 200, 200, 300, 400, 400, 600

Nish believes that there are 5 outliers at the lower end of the data, which gives the range of possible values for x as $1267 \leqslant x < 1400$

Given that there are only 5 outliers **in total** in the raw data

(c) explain why Nish's belief is **not** correct.

(2)

Luke wants to investigate the Daily Mean Windspeed (Beaufort conversion) from the large data set for Leeming in 2015.

He plans to select data until he has 25 pieces of data from **each** of the three categories of windspeed (light, moderate and fresh).

(d) State the sampling technique Luke plans to use.

(1)

(e) Using your knowledge of the large data set, explain clearly why Luke's sampling technique may **not** be successful in obtaining his sample.

(1)

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Question 5 continued

(Total for Question 5 is 9 marks)

TOTAL FOR STATISTICS IS 30 MARKS